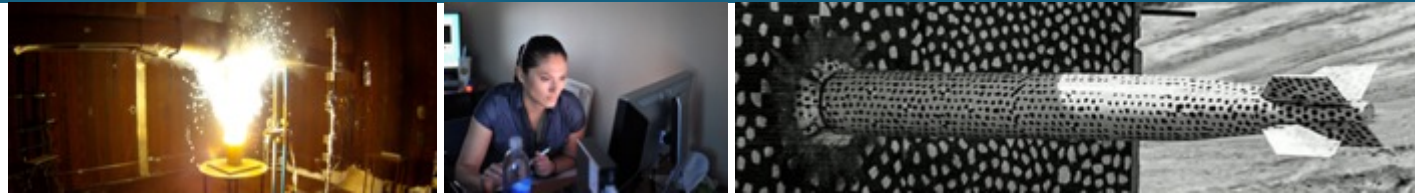




Sandia
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Not So Hot Triangulations



International Meshing Roundtable June 2021

Scott A. Mitchell, Patrick Knupp,
Sarah Mackay, Michael F. Deakin

Center for Computing Research, Sandia National Labs
Dihedral, LLC.



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Not So HOT Triangulations — Overview



Mesh optimization for weighted primal-dual meshes

- Better meshes for primal-dual simulations, e.g. electro-magnetism, fluid flow; divergence and curl

“HOT Triangulations” optimized the “HOT” metric directly

- There are issues; HOT optimization is “not so hot”

We define new metrics with barriers

- Not the same as HOT, so “not so hot,” but optima are qualitatively similar

We do discrete changes

- Edge collapse

We optimize

- More reliable and robust to starting mesh and parameter choices

Title = triple entendre

What is a Primal-Dual mesh?

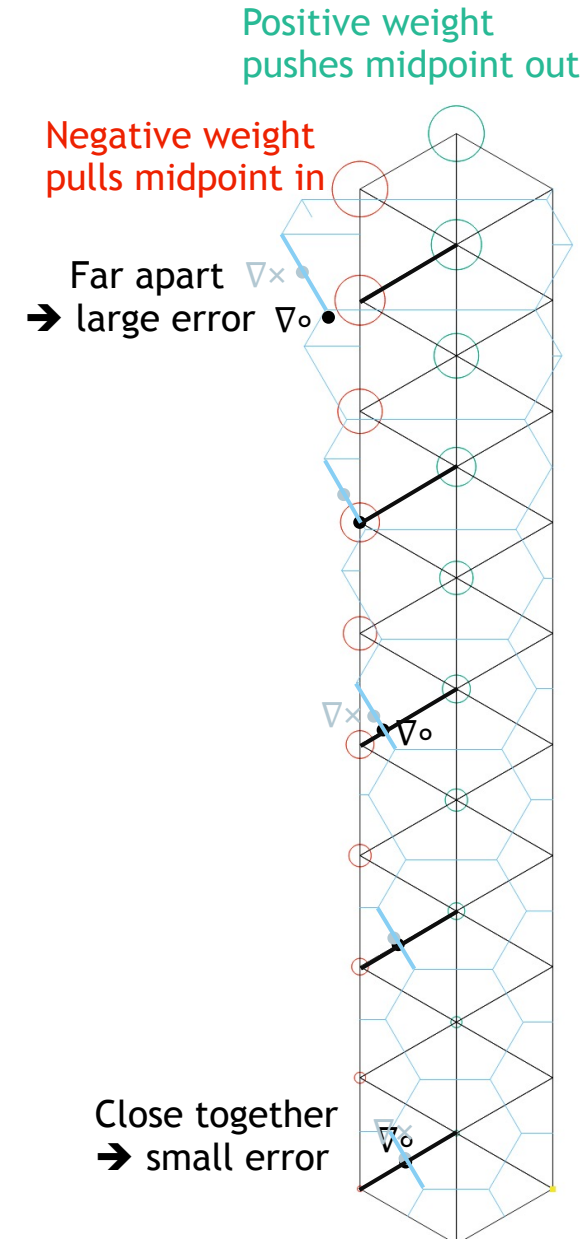


Primal-dual meshes

- Duality
 - Delaunay—Voronoi
 - Regular Triangulations — Power Diagrams
- Shape of primal and dual matters
distance between primal and dual matters
 - $\nabla \circ$ Divergence stored on primal edge midpoint
 - $\nabla \times$ Curl stored on dual edge midpoint
 - Midpoints $\leftrightarrow$, then discretization error $\uparrow$
 - **HOT energy formula** $>$ **discretization error**

Goal: mesh optimization to reduce discretization error

- Optimization of positions and weights
 - **Weight** moves midpoints, additional degrees of freedom

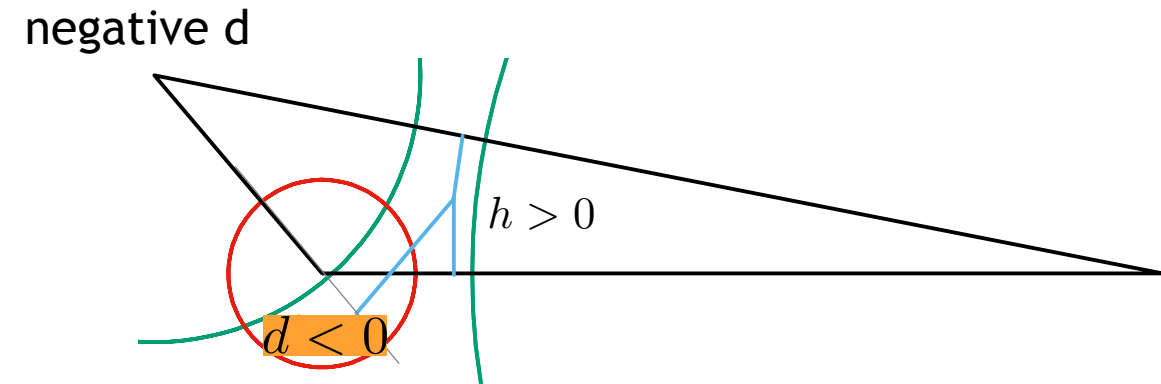
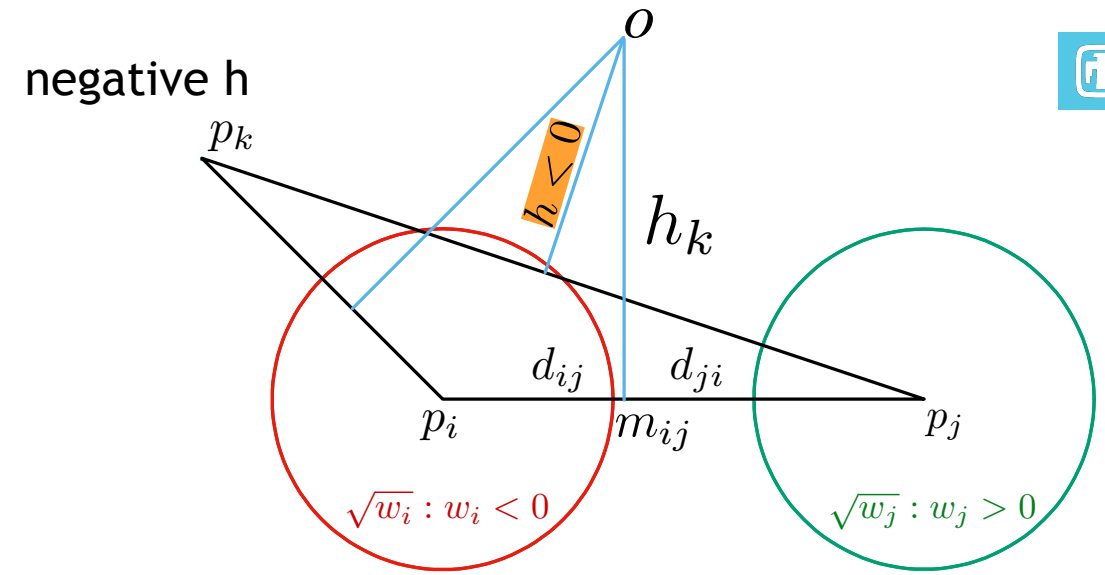
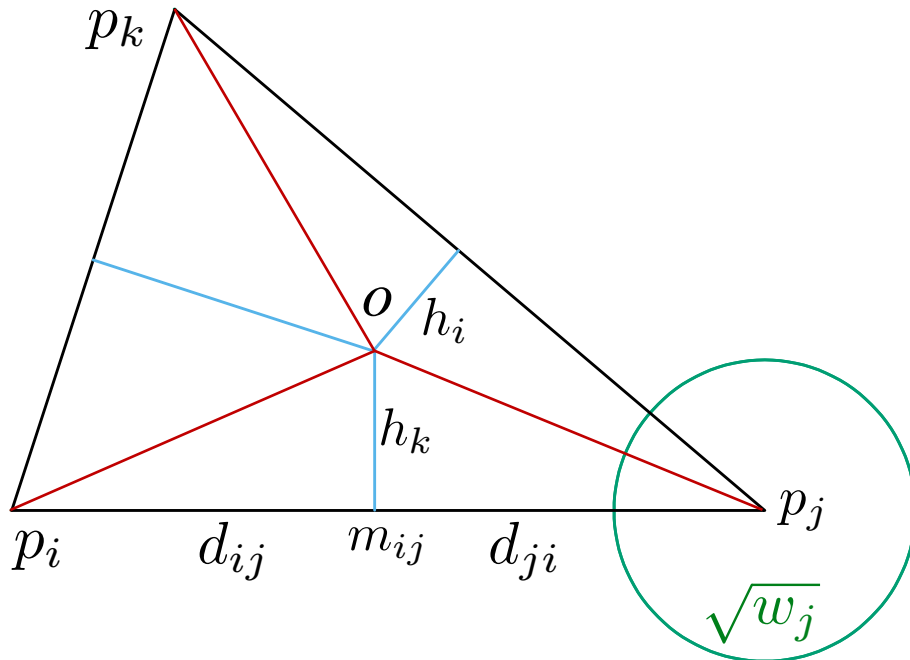


HOT Energy

- “HOT energy” bounds the discretization error for diagonalized Delaunay Hodge star operator
 - HOT = Hodge-Optimized Triangulations

$$\star^s \text{HOT}_{p,p}(\mathcal{M}) = \sum_{\sigma \in \Sigma^s} |* \sigma| |\sigma| W_p(\mu_\sigma, \mu_{*\sigma})^p$$

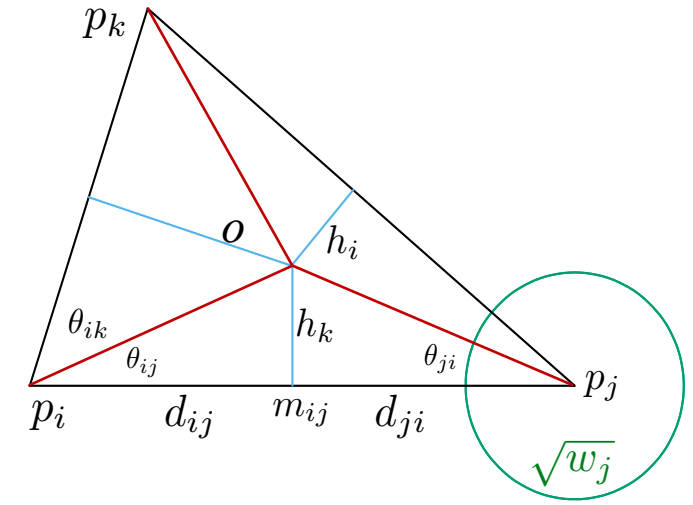
$$\star^1 \text{HOT}(ijk) = d_{ij}^3 h_k + d_{ij} h_k^3$$



Negative h or d →
negative terms in HOT.
These can dominate

Problems with HOT Weight Optimization

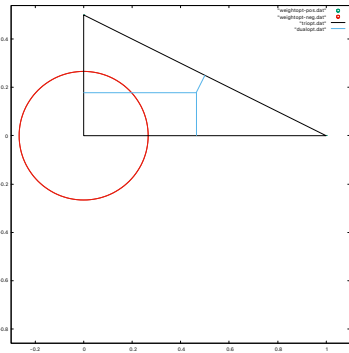
- HOT paper used HOT energy as optimization objective
That approach is “not-so-hot”
 - Weight optimization, fixed positions



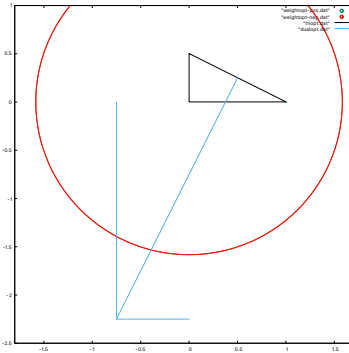
$$\star^1 \text{HOT}(ijk) = d_{ij}^3 h_k + d_{ij} h_k^3$$

Two $h < 0$
 $\text{HOT} < 0$

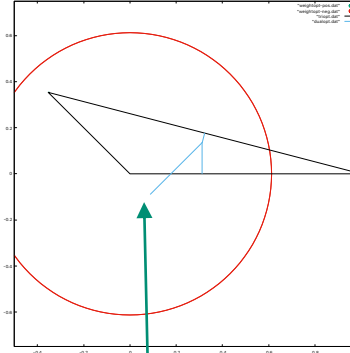
$d < 0$
 $\text{HOT} > 0$



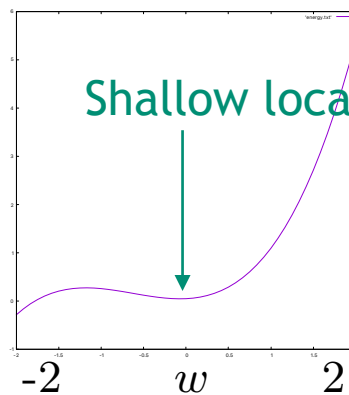
E_F^1 local min



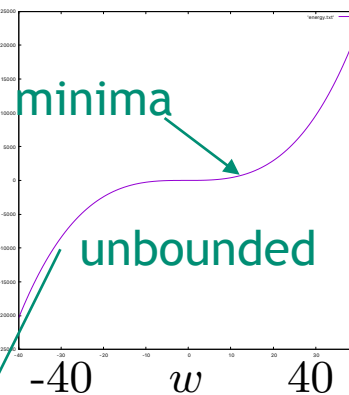
E_F^1 unbounded



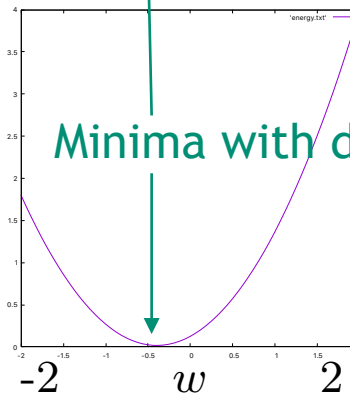
E_F^2 min



Shallow local minima



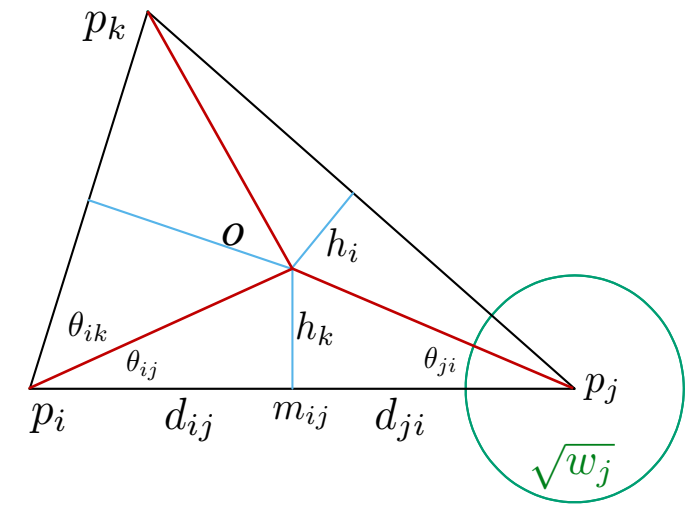
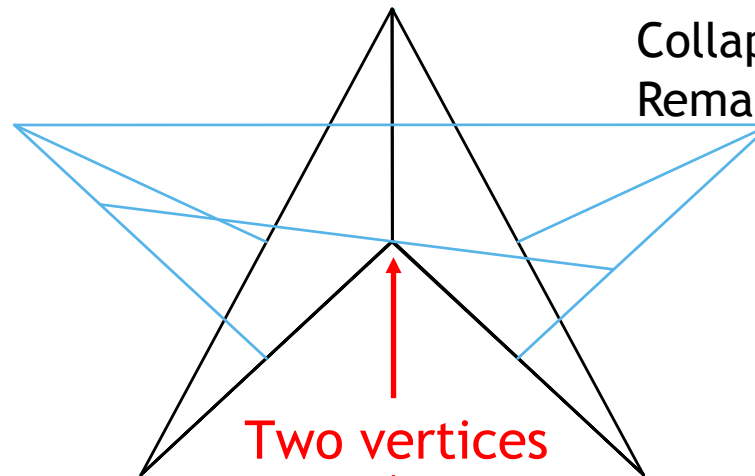
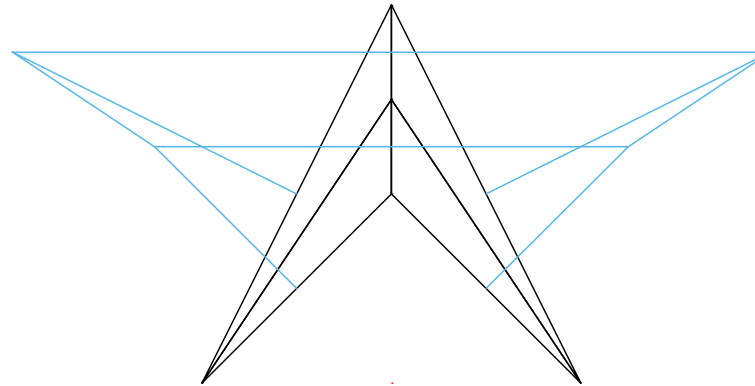
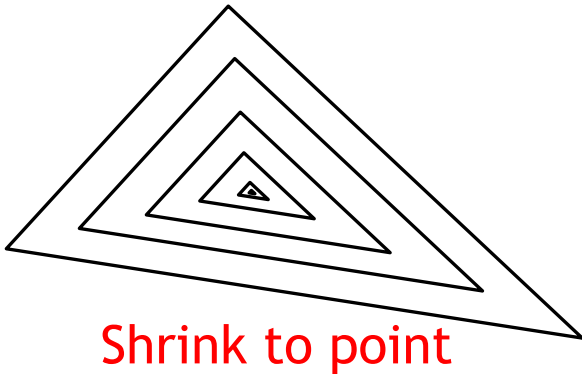
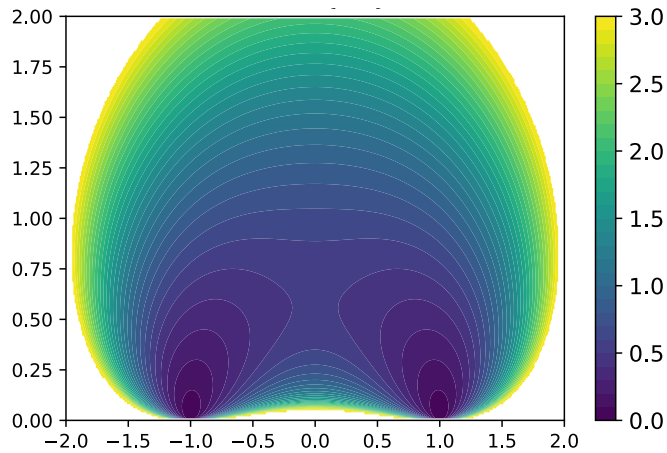
unbounded



Minima with $d_{ij} < 0$, unusable

Problems with HOT Position Optimization

- HOT paper used HOT energy as optimization objective
That approach is “not-so-hot”
 - Position optimization
 - Collapsed elements (or inverted depending on functional form)

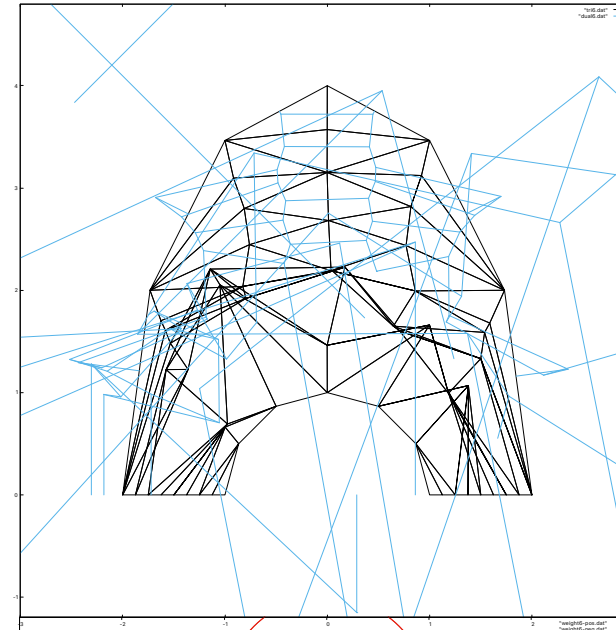
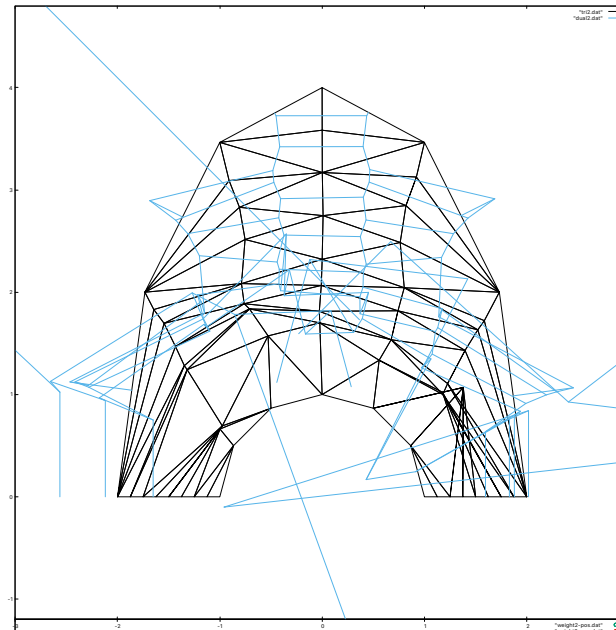
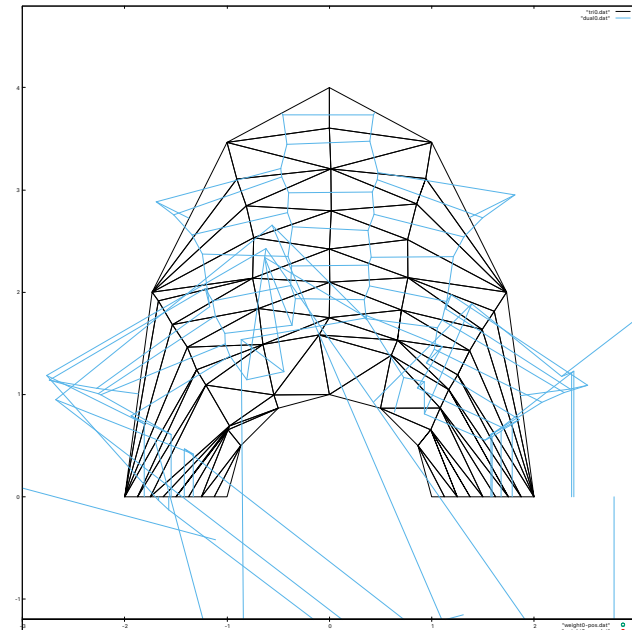


$$\star^1 \text{HOT}(ijk) = d_{ij}^3 h_k + d_{ij} h_k^3$$

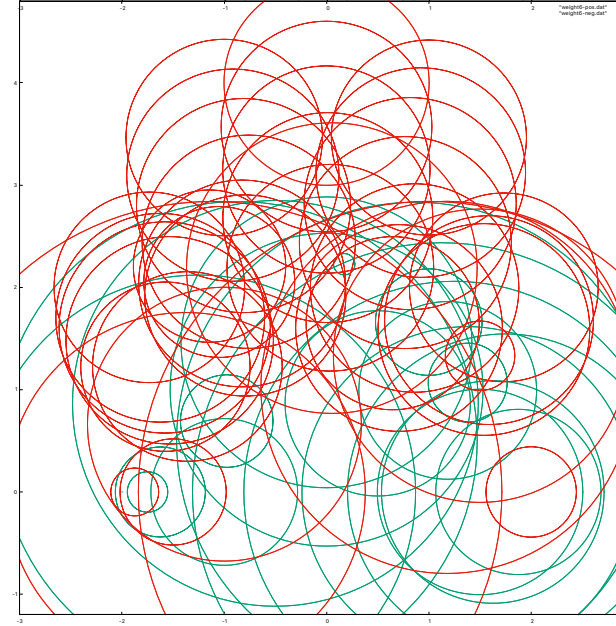
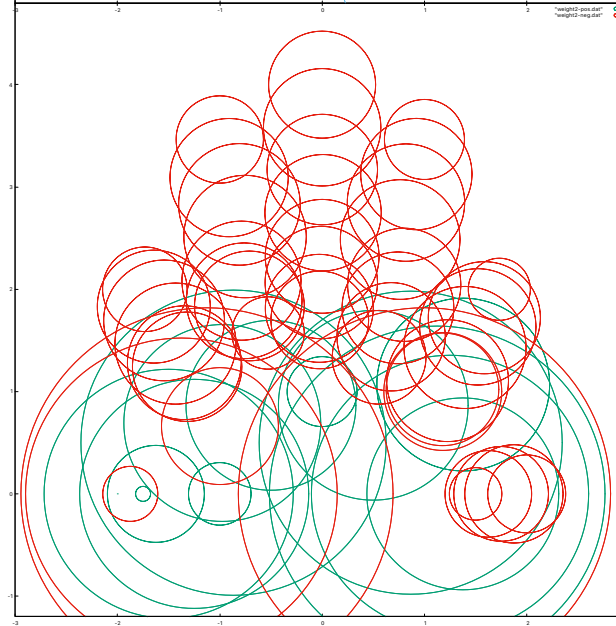
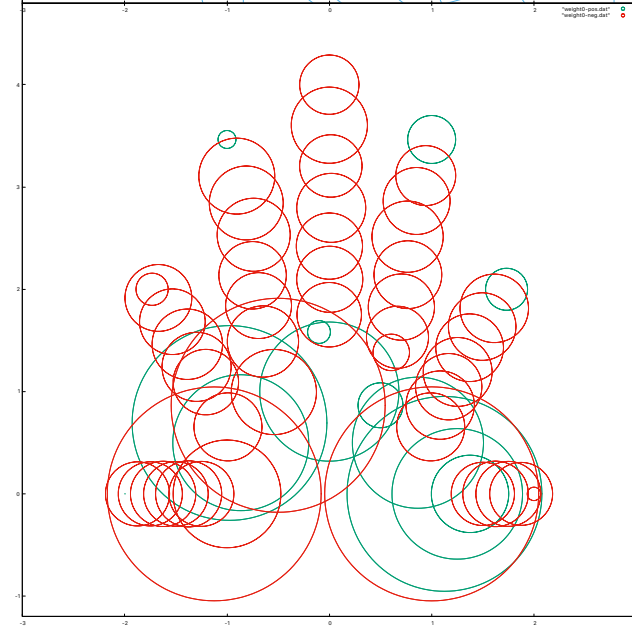
Collapsed triangles have HOT = 0
Remaining triangles are better shaped

Problems with HOT position and weight optimization

“HOT mess!”



Inverted triangles



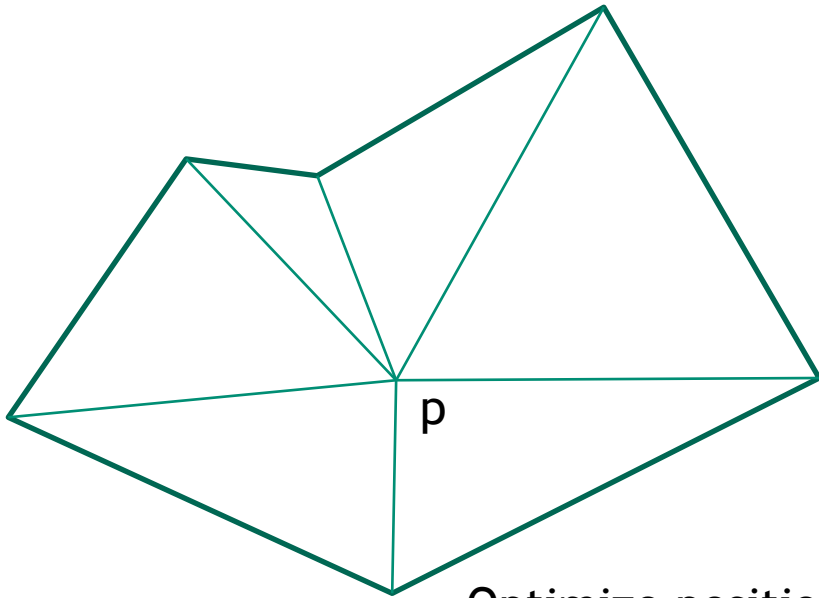
Large weights
large negative energies

Our Solution: Discrete Collapses

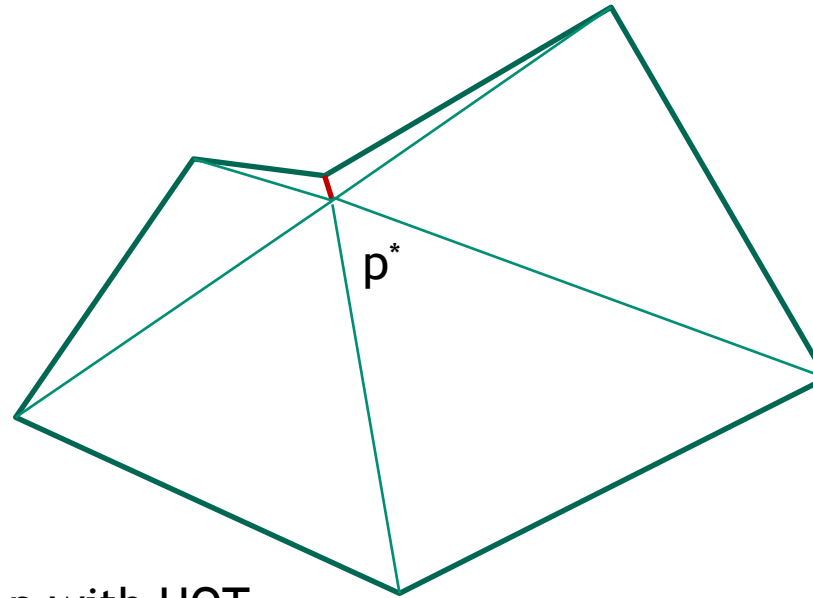
$$\star^1\text{HOT}(ijk) = d_{ij}^3 h_k + d_{ij} h_k^3$$



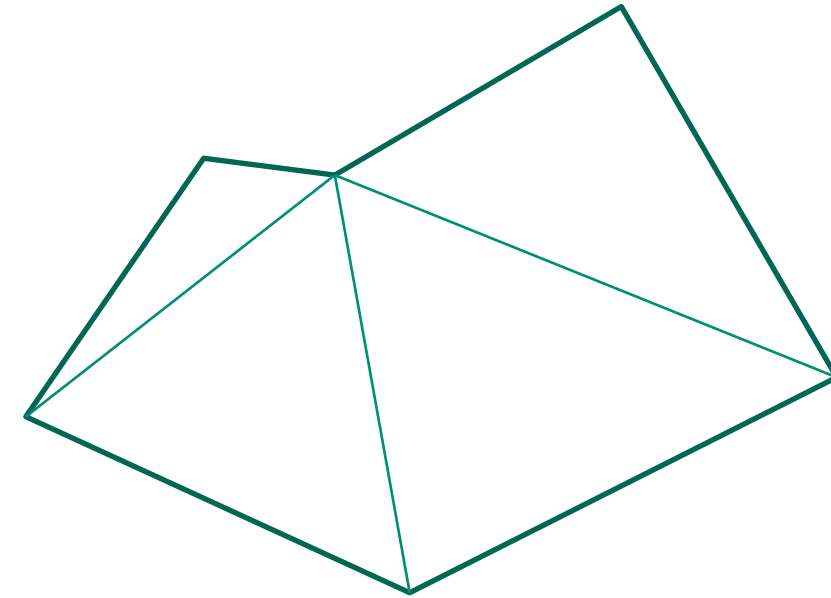
- Collapse vertices: discrete algorithms fix
 - If HOT would collapse or invert, remove vertex instead



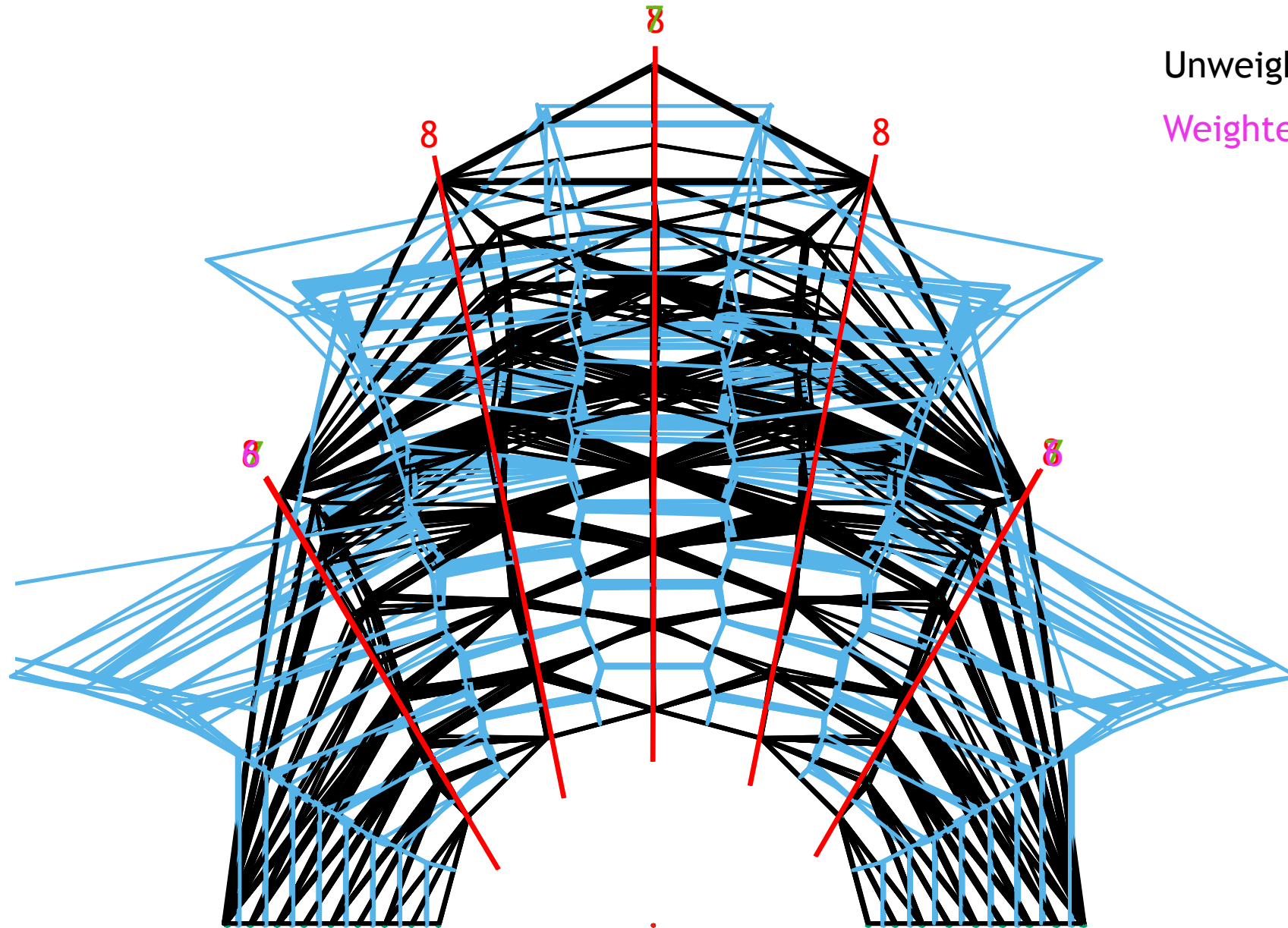
Optimize position p with HOT



If short edge compared to perimeter edge then remove p



Success Story: Opt + Discrete Collapse



Unweighted works = HOT mesh 😊

Weighted unreliable = HOT mess 😞 sometimes

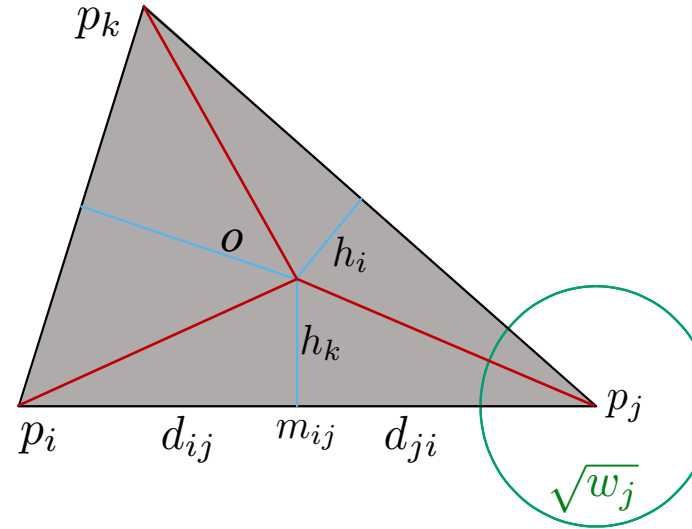
Our Solution: Metrics with Barriers, required when opt weights



$$\star^1 \text{HOT}(ijk) = d_{ij}^3 h_k + d_{ij} h_k^3$$

Contribution of 1 of 6 sub-triangles

- Problems
 - Units length⁴ not scale invariant
 - No barrier to $d_{ij} < 0$
 - No barrier to $h_k < 0$

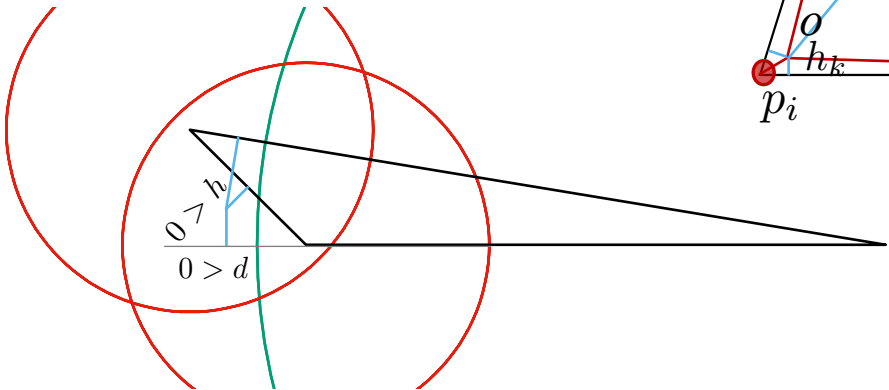
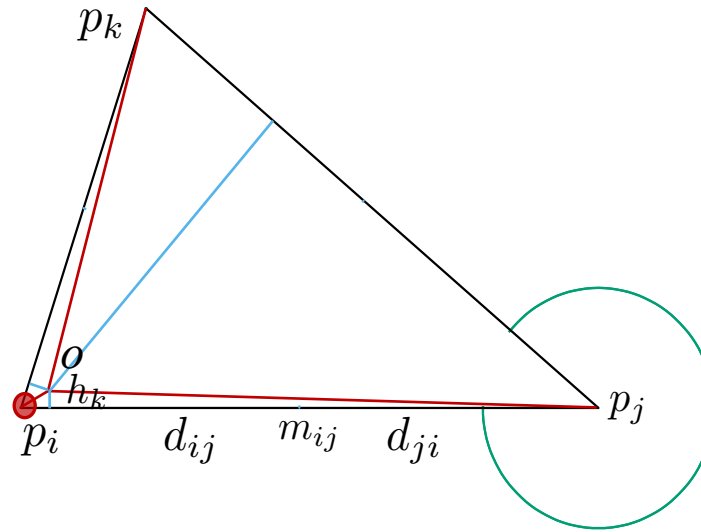


- HOT/area² normalization

- Unitless, scale invariant
- Collapsed triangle $\Leftrightarrow$ Area 0

- HOT / area²($\triangle$) whole triangle

- ✗ Doesn't capture dual, position of o
- ✗✗ No barrier to $d_{ij} < 0$, nor $h_k < 0$



Our Solution: Metrics with Barriers



$$\star^1 \text{HOT}(ijk) = d_{ij}^3 h_k + d_{ij} h_k^3$$

Contribution of 1 of 6 sub-triangles

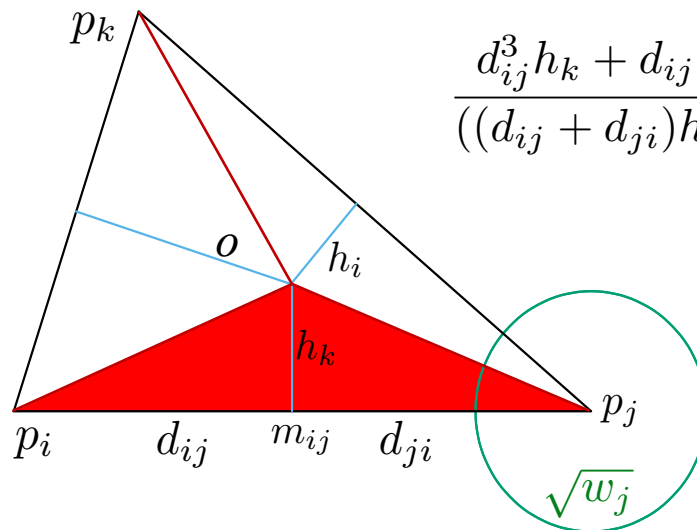
- Problems
 - Units length⁴ not scale invariant
 - No barrier to $d_{ij} < 0$
 - No barrier to $h_k < 0$

- HOT/area² normalization

- Unitless, scale invariant
- Collapsed triangle $\Leftrightarrow$ Area 0

- HOT / area²($\triangle$) 1-of-3 triangle**

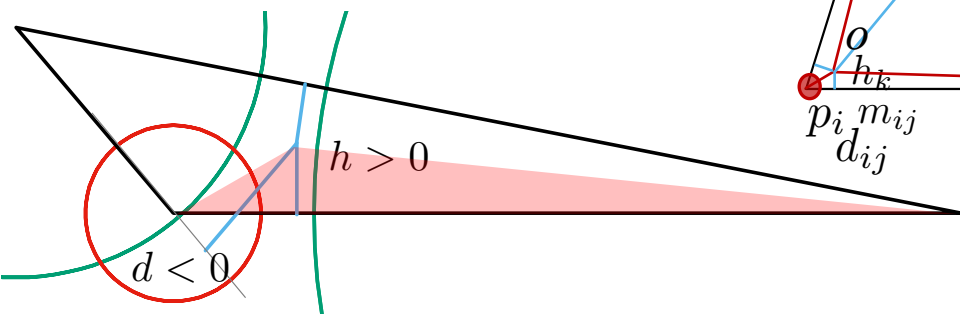
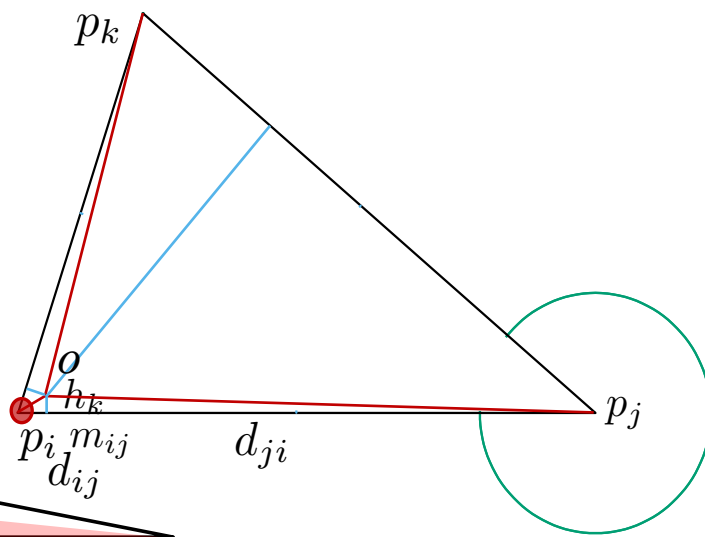
- ✓ Captures dual position of o
- ✓ Barrier to $h_k < 0$
- ✗ No barrier to $d_{ij} < 0$



$$\frac{d_{ij}^3 h_k + d_{ij} h_k^3}{((d_{ij} + d_{ji}) h_k)^2} = \frac{d_{ij}^3}{(d_{ij} + d_{ji})^2 h_k} + \frac{d_{ij} h_k}{(d_{ij} + d_{ji})^2}$$

Barrier to $h_k < 0$

No barrier to $d_{ij} < 0$



Our Solution: Metrics with Barriers

$$\star^1 \text{HOT}(ijk) = d_{ij}^3 h_k + d_{ij} h_k^3$$

Contribution of 1 of 6 sub-triangles

- Problems
 - Units length⁴ not scale invariant
 - No barrier to $d_{ij} < 0$
 - No barrier to $h_k < 0$

- HOT/area² normalization

- Unitless, scale invariant
- Collapsed triangle $\Leftrightarrow$ Area 0

- HOT / area²(Δ) 1-of-6 triangle**

- Captures dual, position of o
- Barrier to $h_k < 0$
- Barrier to $d_{ij} < 0$

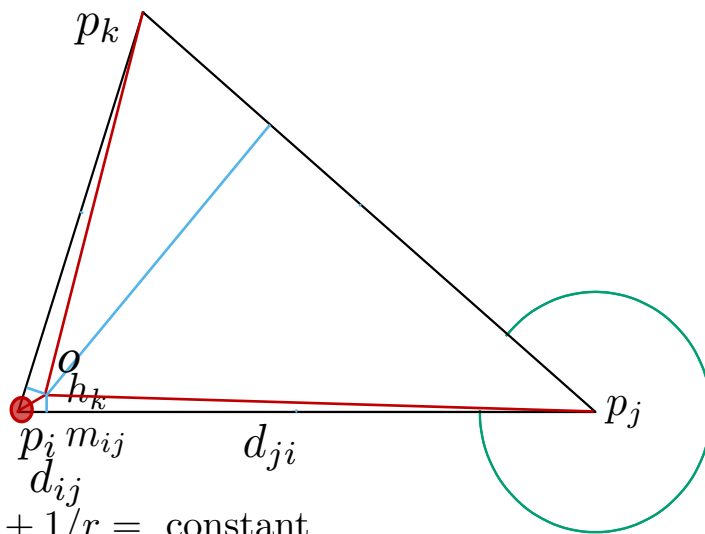
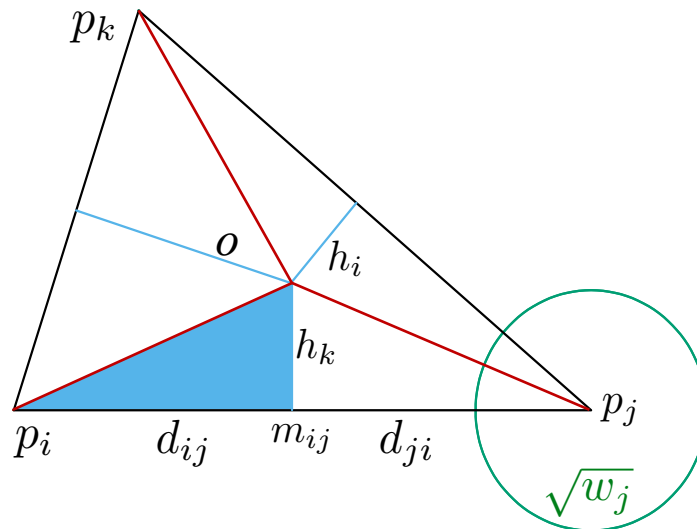
$$h_k \rightarrow 0 \implies E \rightarrow \infty \quad \checkmark$$

$$d_{ij} \rightarrow 0 \implies E \rightarrow \infty \quad \checkmark$$

$$\times \text{ if } \frac{d_{ij}}{h_k} = r \text{ then } d_{ij}, h_k \rightarrow 0 \implies E \rightarrow r + 1/r = \text{constant}$$

$\checkmark$ But that can't happen for all 6 triangles simultaneously (unless $\Delta \rightarrow 0$)

$$\star^s \text{HOT}_{p,p}(\mathcal{M}) = \sum_{\sigma \in \Sigma^s} |* \sigma| |\sigma| W_p(\mu_\sigma, \mu_{*\sigma})^p$$



$$\frac{d_{ij}^3 h_k + d_{ij} h_k^3}{(d_{ij} h_k)^2} = \frac{d_{ij}}{h_k} + \frac{h_k}{d_{ij}}$$

Barrier to $h_k < 0$

Barrier to $d_{ij} < 0$

= condition number of each 1-of-6 triangle
Works well. Not in paper for brevity ...

Our Solution: Metrics with Barriers



- Problem with hard barrier
 - $h_k > 0$ only for acute triangulations in unweighted case
can't start the optimization outside the barrier
- Solution: pseudo-barrier metric E_{BP}
 - Replace h_k with phi in denominator

$$\phi = \frac{1}{2} \left(h_k + \sqrt{h_k^2 + \gamma^2} \right) > 0$$

- But that destroys $d_{ij} < 0$ barrier

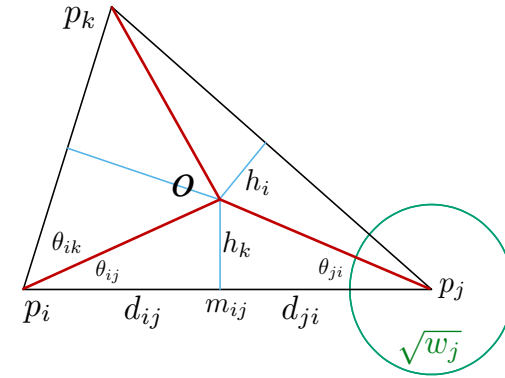
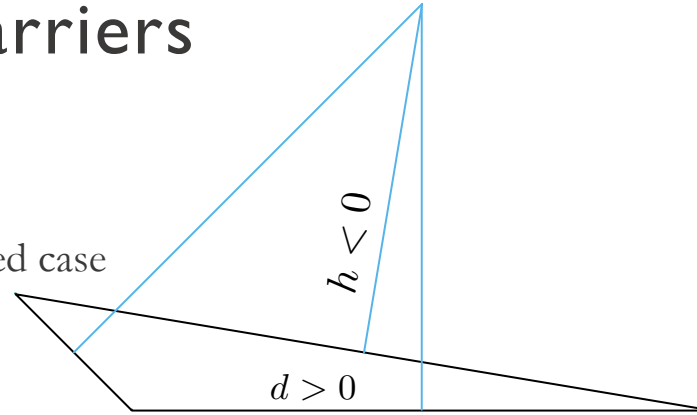
✗ if $\frac{d_{ij}}{h_k} = r$ then $d_{ij}, h_k \rightarrow 0 \implies E \rightarrow r + 1/r = \text{constant}$

✗ And that can happen while other triangles $h_k \rightarrow 0$ and $E \rightarrow 0$

- So replace d_{ij} with $\sqrt{d_{ij}d_{ji}}$ to restore $d_{ij} < 0$ barrier

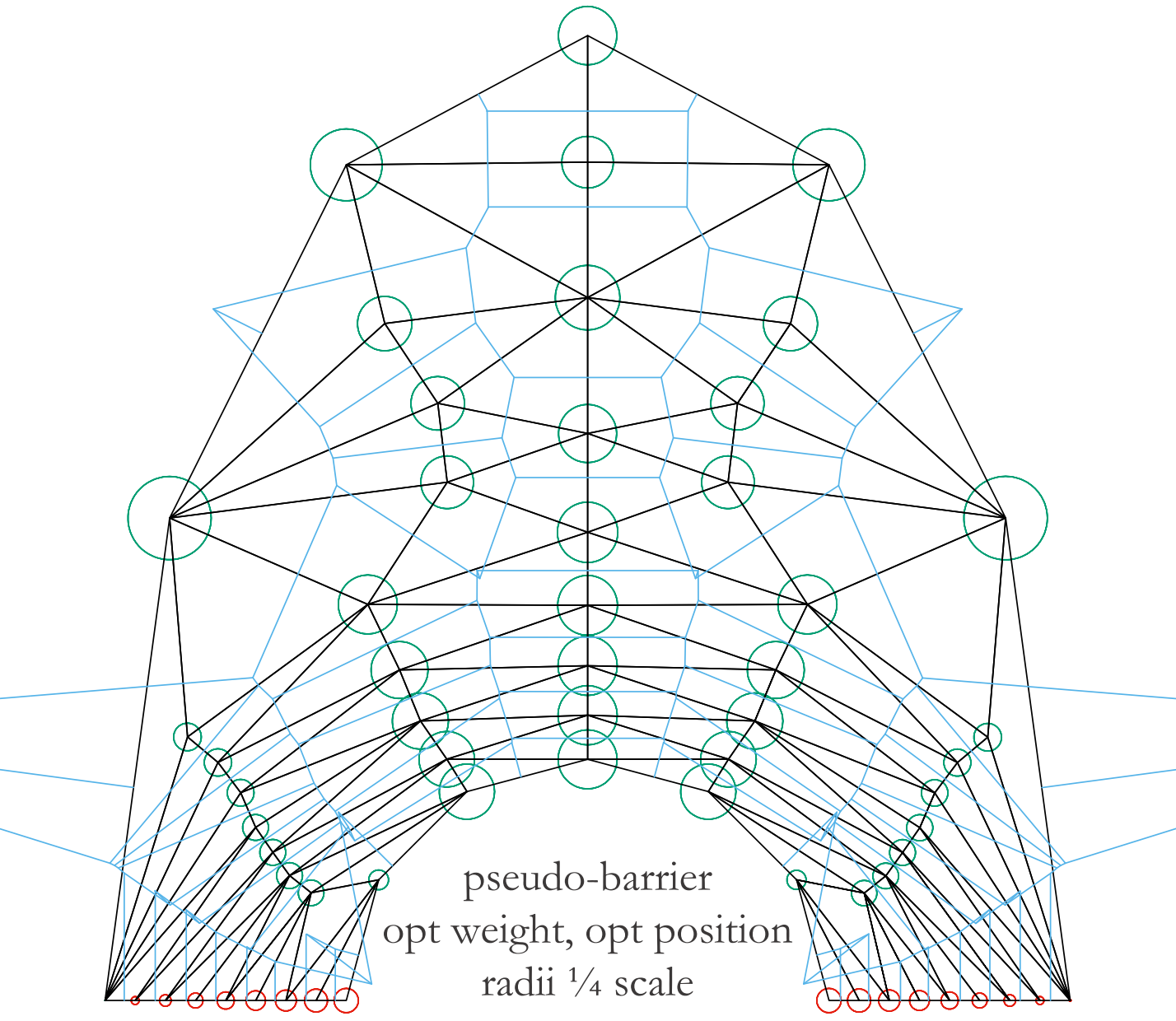
$$E_{BP}(ijk) = \frac{d_{ij}^2 + h_k^2}{\frac{1}{2} \left(h_k + \sqrt{h_k^2 + \gamma^2} \right) \sqrt{d_{ij}d_{ji}}}$$

$$E_{BH}(ijk) = \frac{d_{ij}^2 + h_k^2}{h_k \sqrt{d_{ij}d_{ji}}}$$



$$\star^1 \text{HOT}(ijk) = d_{ij}^3 h_k + d_{ij} h_k^3$$

Barrier-Metric Optimization, position and weights

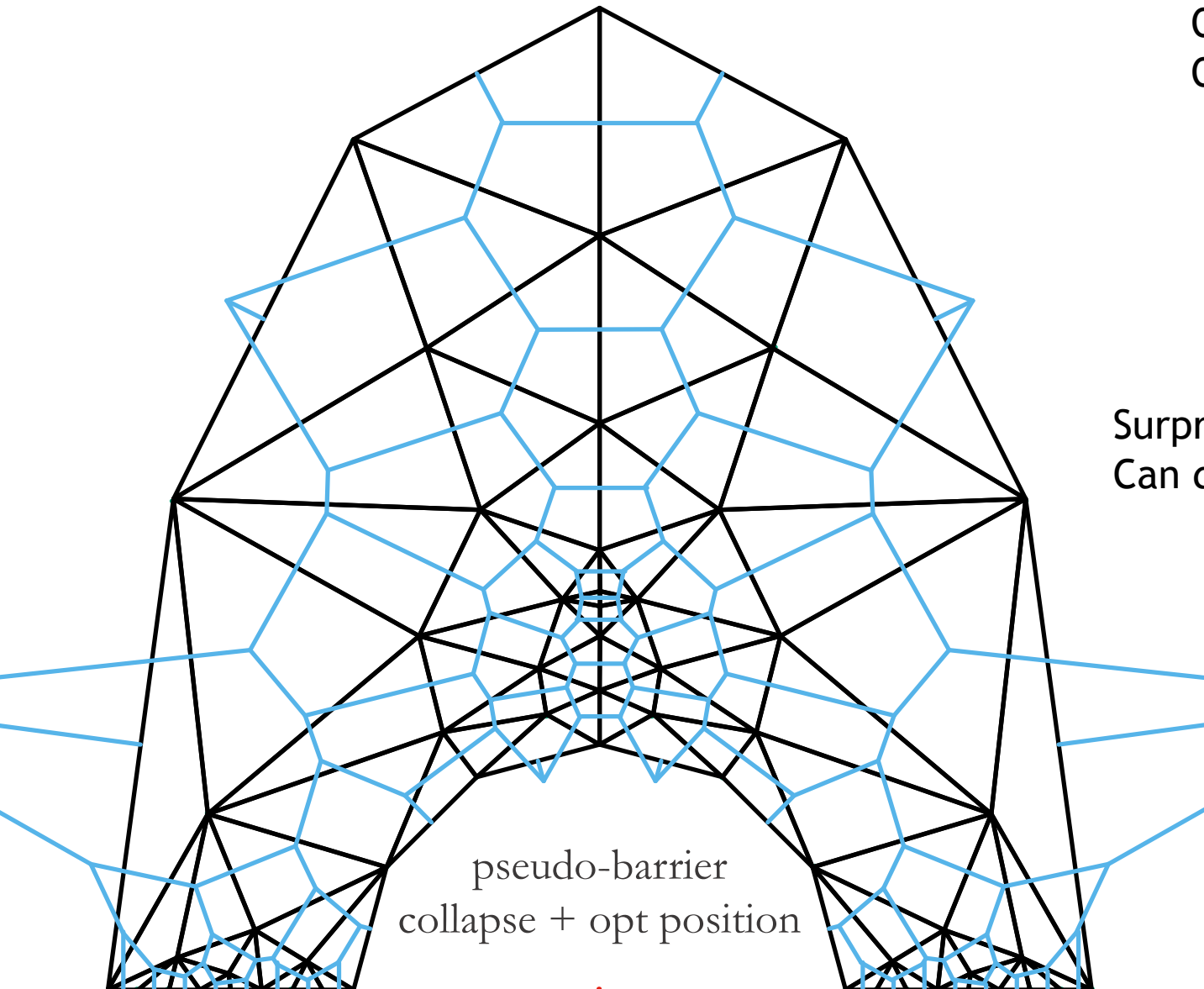


Barrier-Metric Optimization, position and collapses



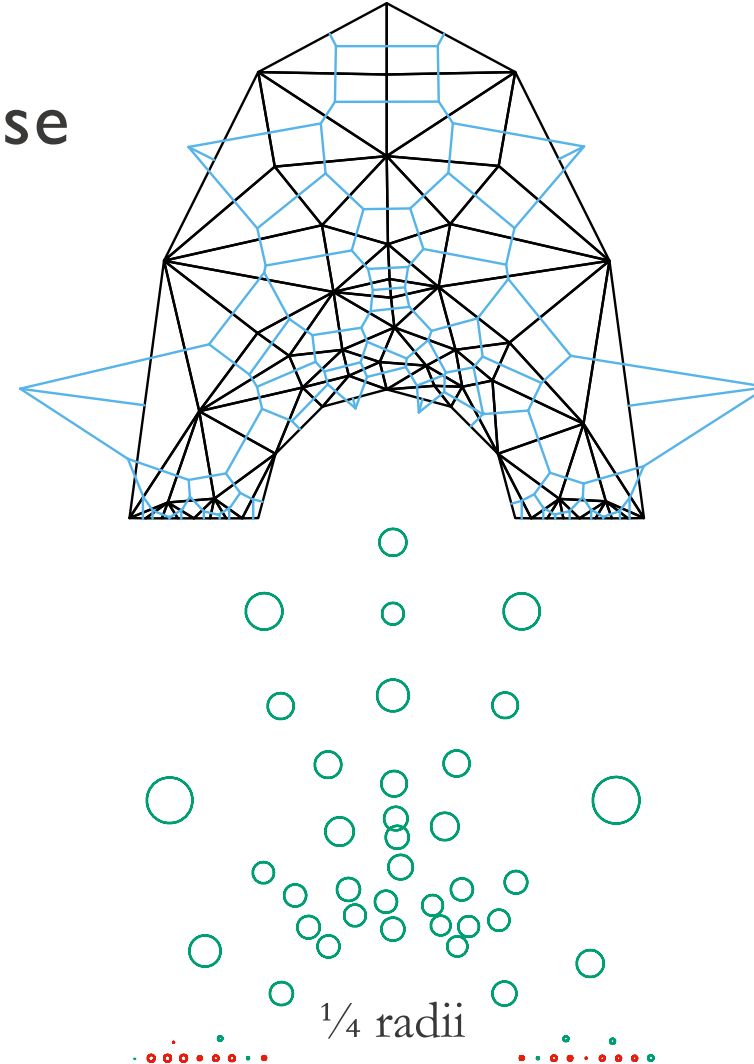
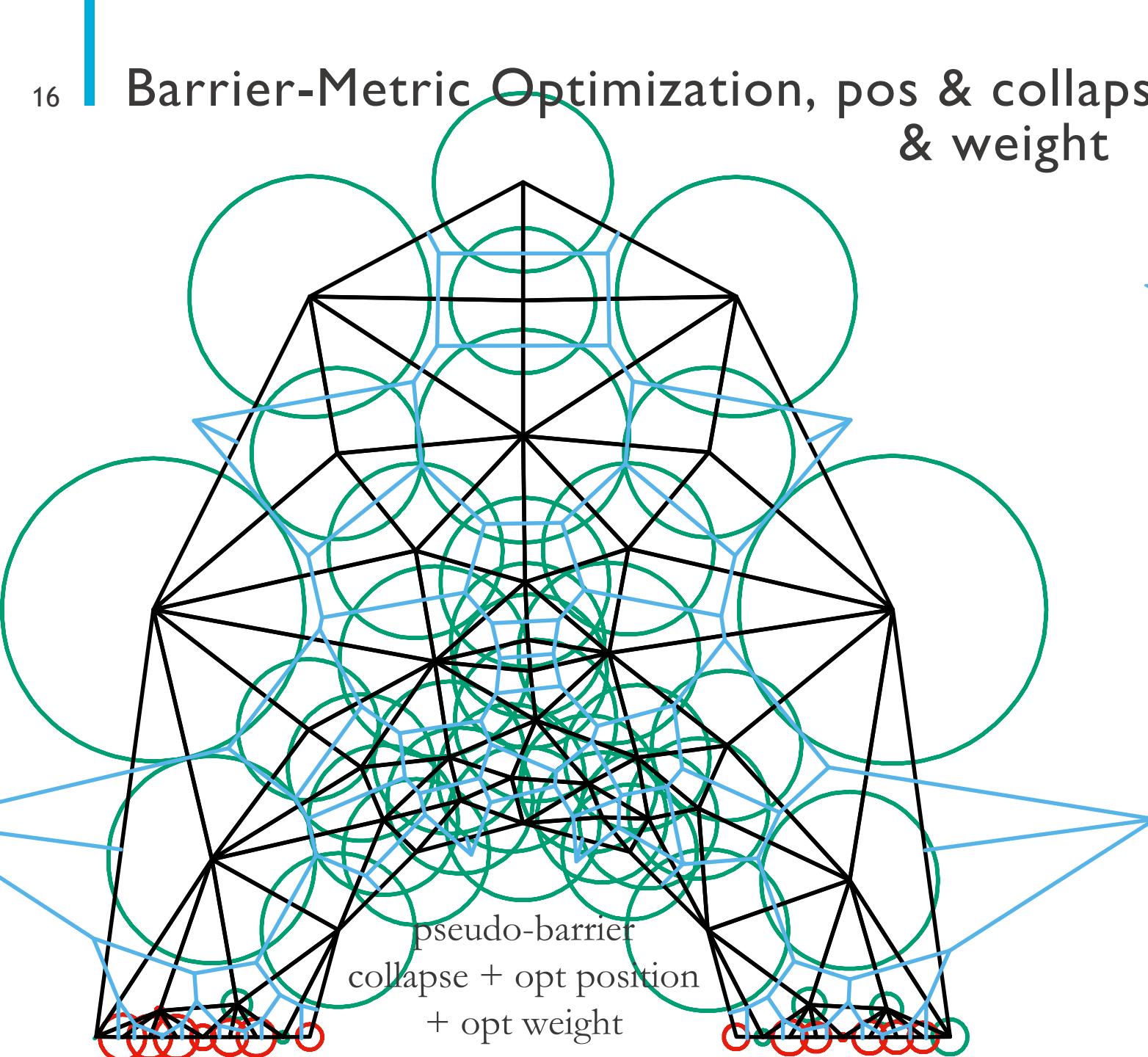
Optimize HOT to collapse
Optimize barrier to collapse
opposite goals

Surprise!
Can do both, alternating optimization steps



pseudo-barrier
collapse + opt position

Barrier-Metric Optimization, pos & collapse & weight



Optimization Algorithm



Algorithm 1 Optimize Triangulation $\mathcal{T}$

```

1: Parameters:
2:  $E_o$  energy to optimize, e.g.,  $E_{BP}$ 
3:  $E_c$  energy for collapses  $\in \{E_F, E_T, E_{\partial T}\}$ 
4:  $i_c$  first iteration to consider collapses
5:  $i_w$  first iteration to optimize weights
6: function OPTIMIZE( $\mathcal{T}$  with  $\partial$ )
7:     repeat                                ▷ outer iterations
8:         repeat                            ▷ inner iterations
9:             for point  $p \in \mathcal{T}$  do
10:                 Gather patch  $P(p)$  with
11:                     ordered perimeter  $\{q_j\}$ 
12:                 if not Collapse( $E_c, P(p)$ ) then
13:                      $p = \text{Optimize position}(E_o, P(p))$ 
14:                      $p = \text{Optimize weight}(E_o, P(p))$ 
15:                 shift weights so  $p_{\text{anchor}}$  has weight = 0
16:                 increment iteration  $i$ 
17:             until  $E(\mathcal{T})$  converged and no collapses
18:             Flip  $\mathcal{T}$  to restore regularity
19:     until no flips were needed

```

Problems with non-Regular Triangulations



Hodge operator is only defined for Regular (Delaunay) triangulations

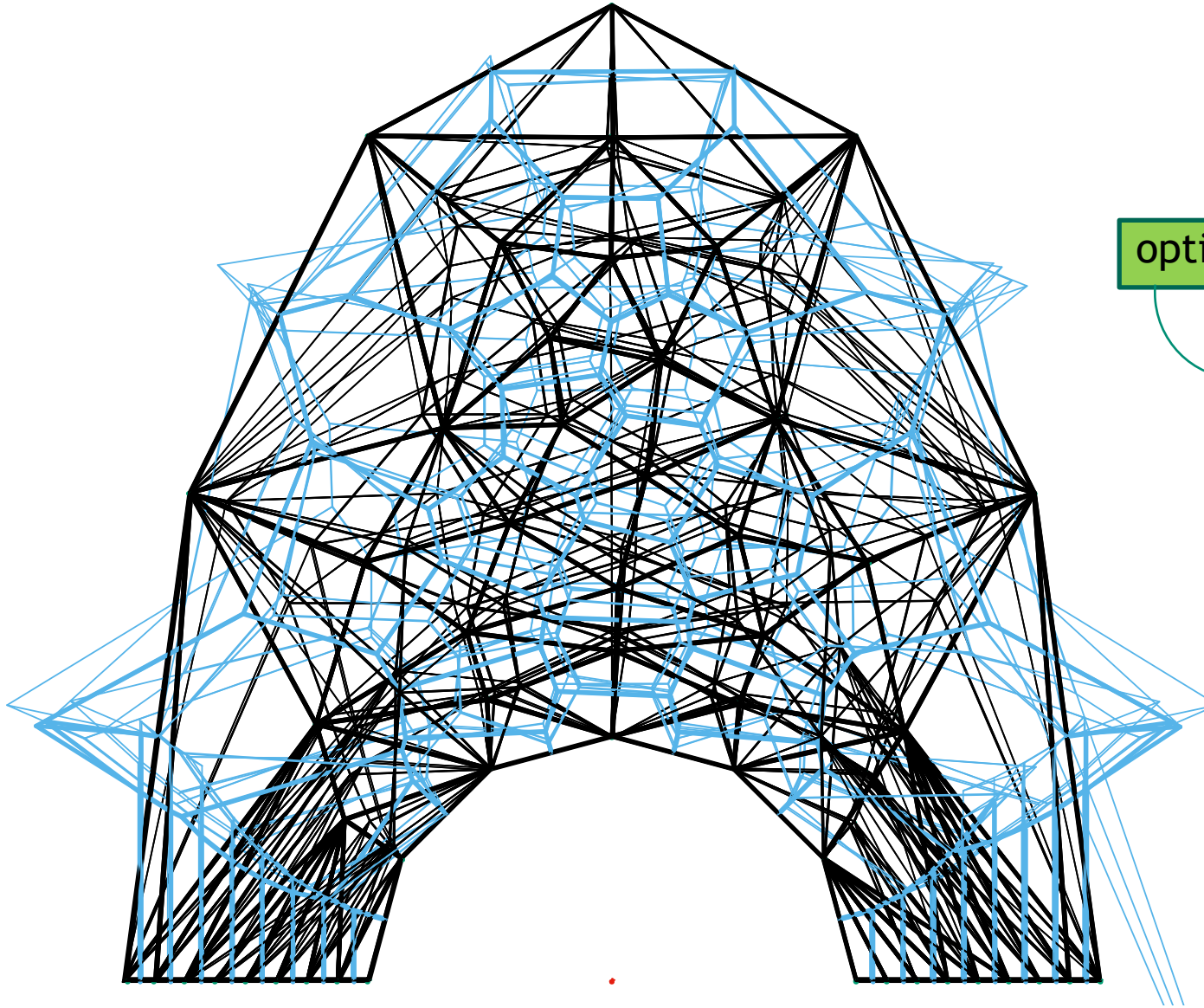
Final mesh **must** be flipped to restore regularity

- E.g. it needs a flip if a dual edge has negative length

If the mesh needs flips, then any Energy is an **extrapolation**

- HOT paper showed HOT energy is a **continuous** extrapolation
- We propose other extrapolations that avoid some unbounded solutions, are more accurate, but not perfect

Opt + Discrete Collapse + Flips and Repeat



optimal but not regular

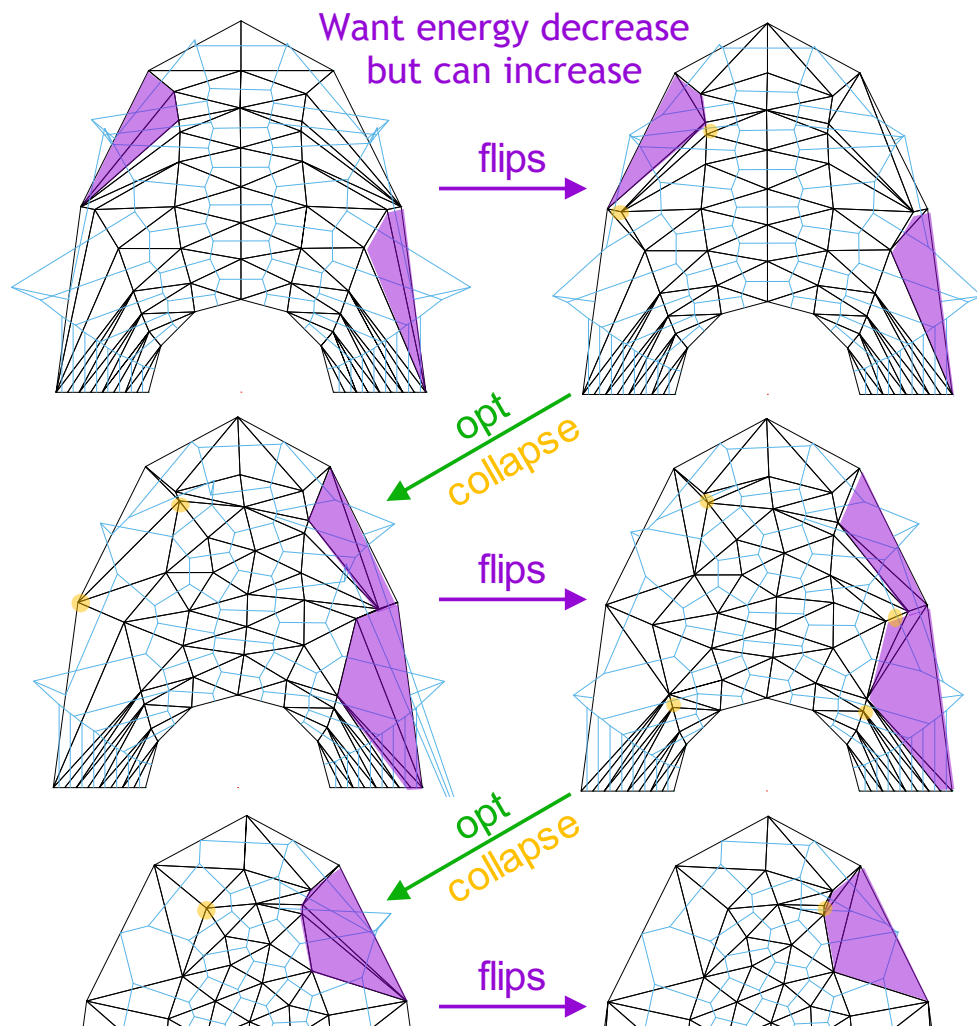
regular but not optimal

Outer loop Optimization after Restoring Regularity

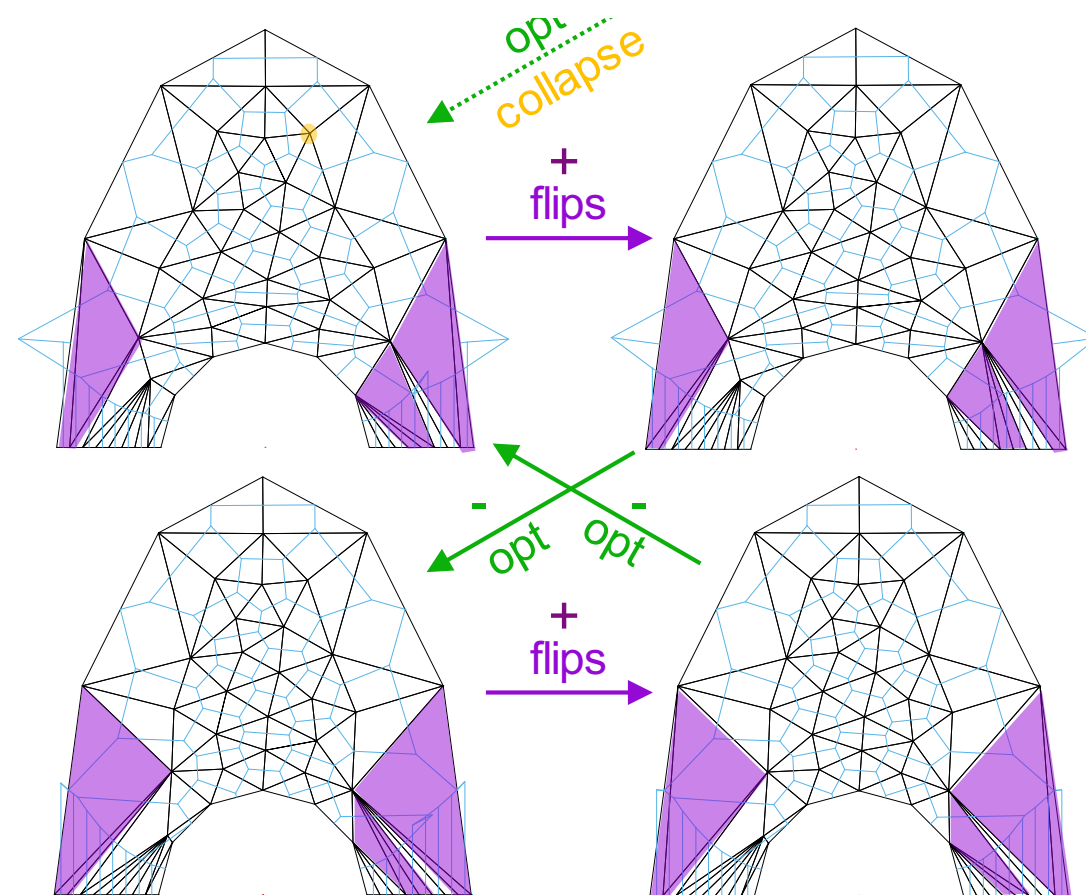


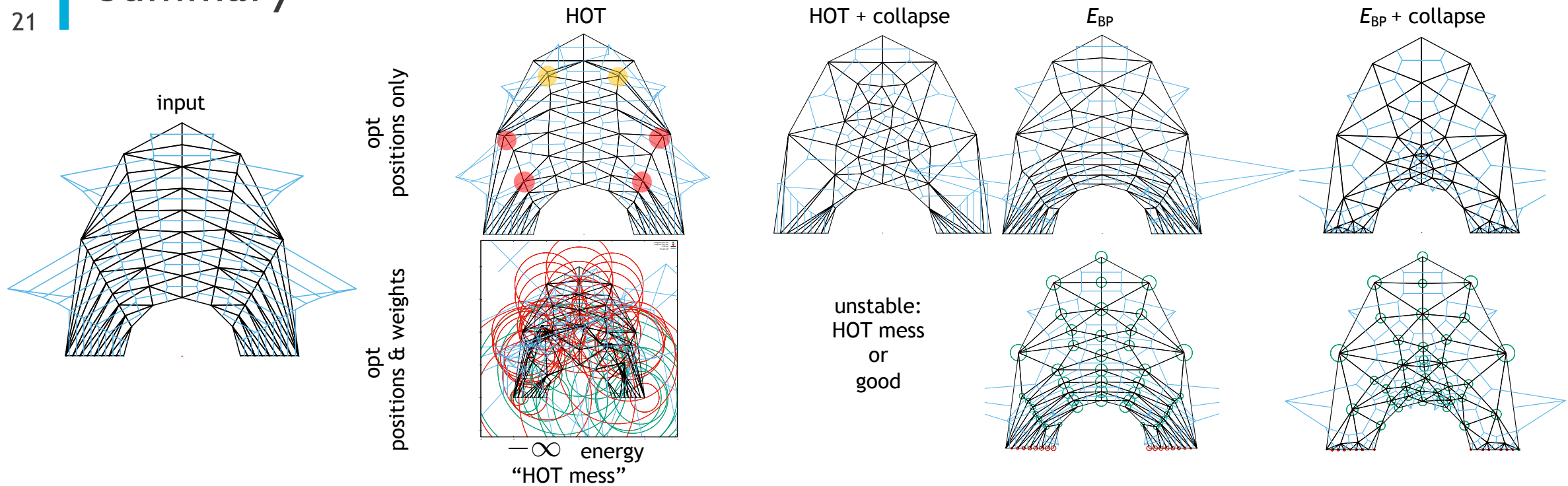
Inaccurate extrapolation:

Energy (HOT, or our energy) can go **up** after flips

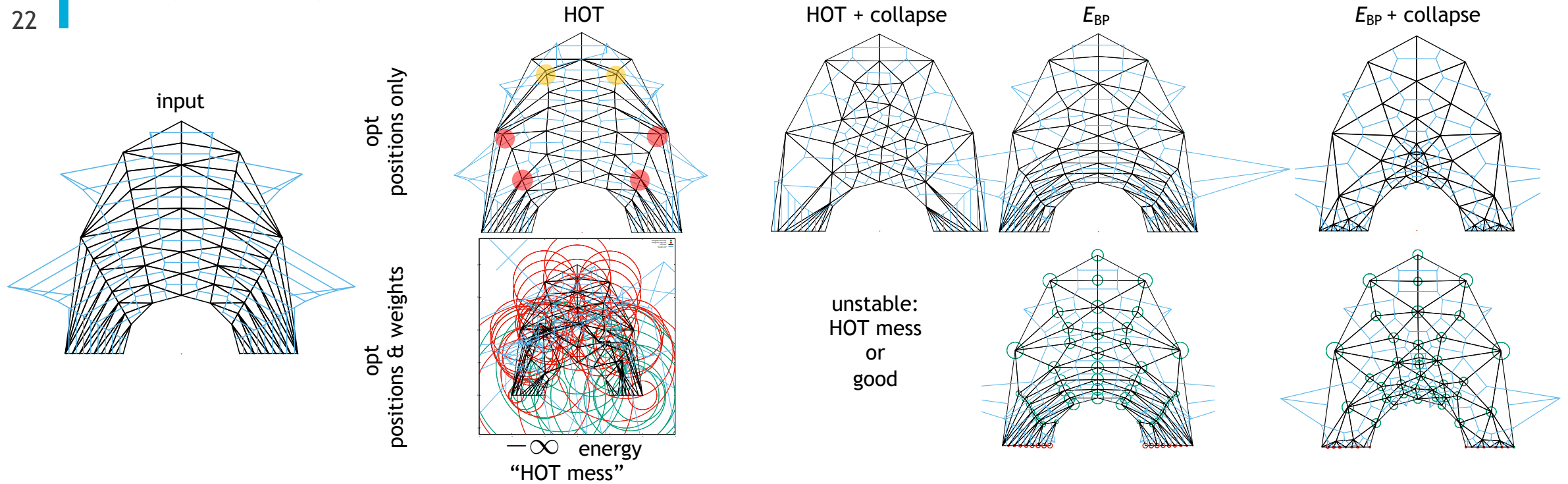


Opt decreases energy
Collapse can increase or decrease





- Inner optimization
 - Collapsing is sufficient for optimization to be reliable if no weights
 - Delaying weight optimization speeds convergence, better meshes, but not enough to make it always reliable
 - Pseudo-barrier metrics are enough to make optimization reliable
 - Best is all three: barrier metrics, collapses, weight optimization after iteration 4
- Outer optimization (re-optimize after flips restore regularity)
 - Improves final mesh
 - Cycles possible, need termination criteria



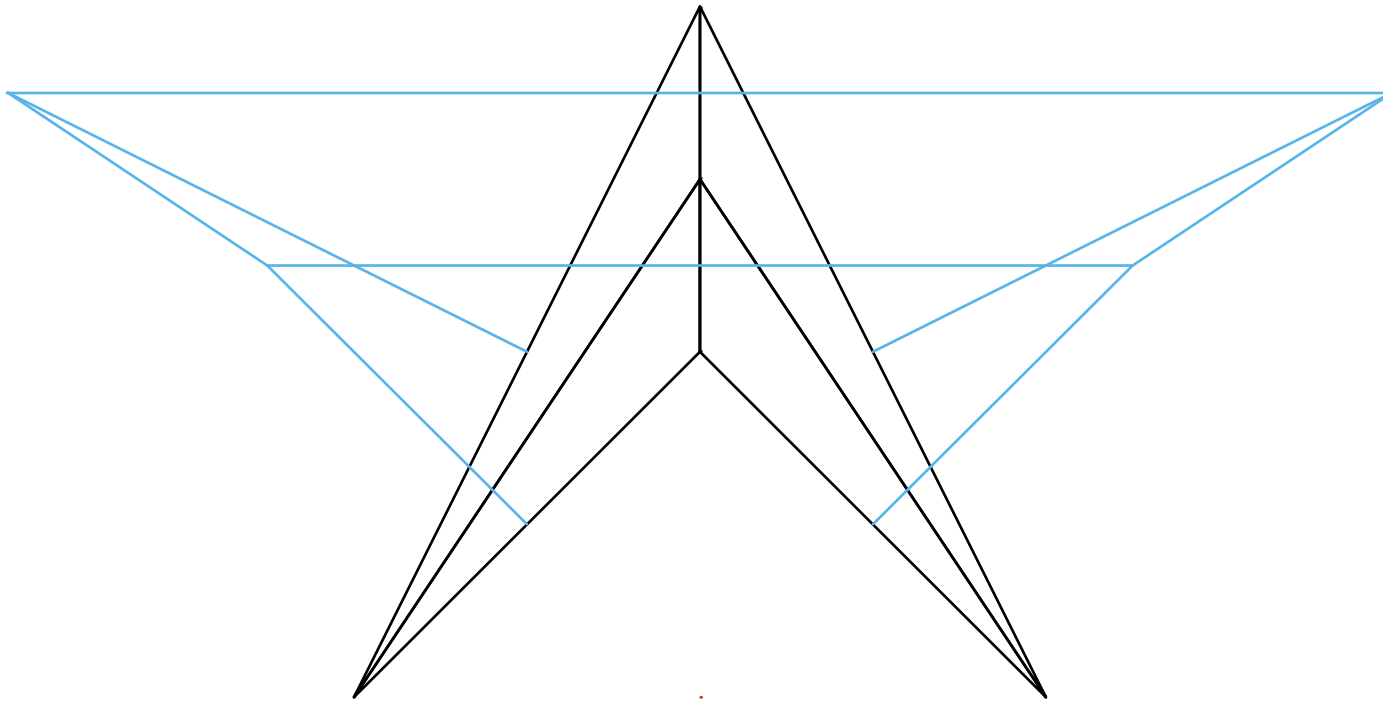
- Future work
 - Additional collapsing rules
 - Insertion rules
 - Dual motivated rules
 - Lots of algorithm parameters and options
 - What's the best mix?

Open Problems



Discover an algorithm that can find a set of weights so all $d_{ij} > 0$ and $h_k > 0$, if any

- Could then optimize the hard barrier metric
- No such weights exists for this mesh:



Primal-Dual Metrics for Different Operators



Is HOT what we should be optimizing?

- HOT comes from the diagonalized Delaunay Hodge star operator from Discrete Exterior Calculus

Mimetic formulation

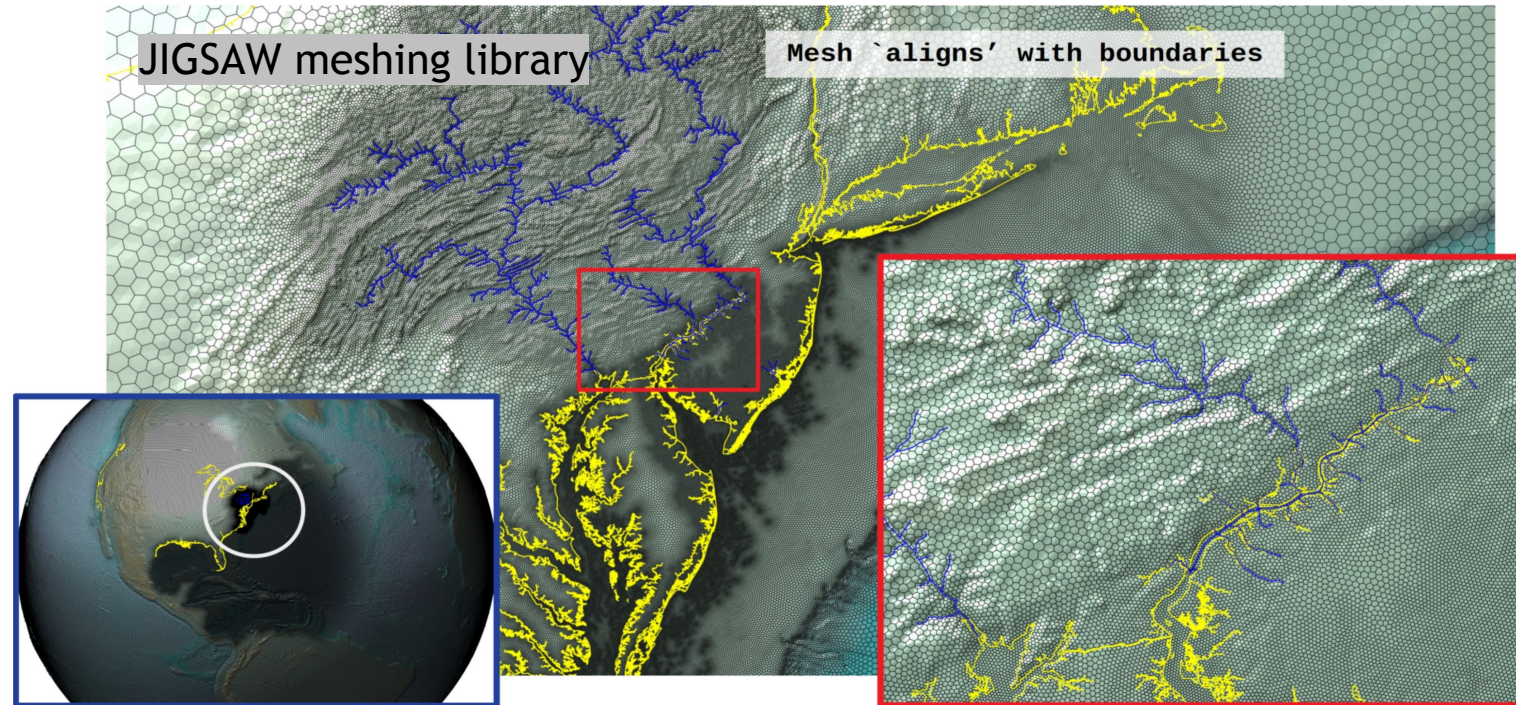
Darren Engwirda

- Different discretization error than HOT
- Objective: distances between midpoints, centroids and orthocenters
- “Tight bounds on the difference and relative difference of the circumradius and orthoradius of a weighted triangle”

Michelle Hummel, this IMR

Alternative to mesh optimization

- Change discretization to match the mesh



‘Unified’ land-river-ocean mesh, ICoM project, Engwirda et al, 2020